Chapter 12: Atoms

Formula Sheet + Tips & Tricks

Formula Sheet

I. α-Scattering Experiments

$$r_0 = \frac{1}{4\pi\varepsilon_0}\frac{Z_1 Z_2 e^2}{E}$$

r_0 = distance of closest approach, Z_1, Z_2 = charge numbers, E = kinetic energy of incident particle

II. Radii of Orbits and Velocities

$$r_n = n^2 r_1, \quad r_1 = 0.529\ \text{Å}(\text{for hydrogen, } Z = 1)$$

r_n = radius of nth Bohr orbit; radius scales as n^2/Z and inversely with orbiting particle's mass

II. Radii of Orbits and Velocities

$$v_n = \frac{e^2}{2\varepsilon_0 n h}$$

v_n = orbital speed in nth orbit; independent of the orbiting particle's mass

II. Radii of Orbits and Velocities

$$T = \frac{2\pi r_n}{v_n}$$

T = time period of revolution in the nth orbit

III. Energy of Electron in Different Orbits

$$E_n = -\frac{13.6\, Z^2}{n^2}\ \text{eV}$$

E_n = total energy in nth orbit; for a Bohr orbit, $KE = -E_n$ and $PE = 2E_n$

IV/V. Transitions, Excitation and Ionisation

$$\Delta E = E_i - E_f = h\nu = \frac{hc}{\lambda}$$

ΔE = photon energy absorbed/emitted; ν = frequency, λ = wavelength

V. Excitation and Ionisation Potentials

$$E_{\text{ionisation}} = -E_1 = 13.6\,Z^2\ \text{eV}$$

Ionisation energy = energy to take electron from ground state ($n = 1$) to $n = \infty$; numerically equal to ionisation potential in volts

VI. Series in Hydrogen Spectrum

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

R = Rydberg constant; n_1 = lower level (series), $n_2 > n_1$ = upper level

VI. Series in Hydrogen Spectrum

$$E_{\text{ionisation}} = Rhc$$

Series-limit (ionisation) energy obtained by putting $n_1 = 1,\ n_2 \to \infty$

Tips, Tricks & Hacks

- Rutherford's α-scattering: most particles pass straight through (atom is mostly empty space), a few deflect at large angles (dense, positively charged nucleus) — know which observation supports which conclusion, it's asked as a match-the-following.
- Bohr radius scales as $r_n \propto \frac{n^2}{Z}$ and orbital speed as $v_n \propto \frac{Z}{n}$ — as n increases, orbits get bigger and slower; as Z increases (heavier hydrogen-like ion), orbits shrink and speed up.
- Energy of electron in nth orbit: $E_n = \frac{-13.6\,Z^2}{n^2}$ eV — memorise 13.6 eV for hydrogen ground state, then just scale by Z^2/n^2 for any hydrogen-like ion or excited state.
- Energy released absorbed on transition $n_2 \to n_1$ uses $\Delta E = 13.6Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$ eV — for emission $n_2 > n_1$ (energy released, photon emitted); for absorption $n_1 < n_2$ needs energy input.
- Ionisation energy is the energy to take an electron from its current level to $n = \infty$ (i.e., $E = 0$); excitation energy is between two *bound* levels — don't confuse the two when a question says "minimum energy to ionise" vs "minimum energy to

excite to next level".

- Spectral series memory: Lyman (UV, ends at $n = 1$), Balmer (visible, ends at $n = 2$), Paschen/Brackett/Pfund (IR, end at $n = 3, 4, 5$) — "Lyman is Lowest ($n = 1$), Balmer is visible (the one you can actually see)" is a quick recall hook.
- Maximum number of spectral lines from an electron de-exciting from level n is $\dfrac{n(n-1)}{2}$ — a very frequently tested combinatorics-style formula.